IBM University Engagement Project Submission

"Exploring the Power of Agentic AI with IBM Granite and LangFlow"

Project Tile - Intelligent Shopping Agent

Domain of Project - E-commerce

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Problem Statement -

Problem Statement : Intelligent Shopping Agent

The Challenge

Shoppers today face an overload of product options, scattered reviews, and dynamic pricing across e-commerce platforms. Finding the right product that balances price, quality, and personal preferences is time-consuming and frustrating. Traditional recommendation systems often fail to capture real-time trends, personalized needs, or contextual factors like urgency, budget, or sustainability. The challenge is not just to recommend products but to act as a smart shopping companion that helps buyers make informed and confident decisions.

The Objective

Build **an agentic AI shopping assistant** that provides intelligent, personalized, and real-time support. The solution should:

- **Fuse Multisource Data :** Aggregate product details, reviews, ratings, and price comparisons across multiple e-commerce sites into unified insights.
- **Enable Multimodal Search :** Allow users to input voice queries, images (e.g., a photo of a product), or text descriptions to find best-matching products.
- **Create a Predictive & Agentic Layer :** Suggest purchase timing (e.g., discounts, seasonal offers), recommend alternatives, and alert users about price drops or fake reviews.
- **Visualize Smart Shopping Pulse :** Provide a dashboard summarizing comparisons, sustainability scores, trending products, and personalized shopping lists.

Proposed Solution -

The proposed solution **Intelligent Shopping Agent** stands out by combining **LangFlow, IBM watsonx.ai,** and **Granite Models** into a single intelligent system. It is developed o act as a smart shopping companion that helps buyers make informed and confident decisions. The system addresses challenges in finding the right product that balances price, quality, and personal preferences.

The system consists of following key agents:

- Multimodal Search Agent - Processes text, voice, and image inputs to identify and match products accurately.
- Shopping Companion Agent - Understands user needs and coordinates all shopping-related tasks and recommendations.
- Product Discovery Agent - Finds the most relevant products across multiple e-commerce platforms based on user preferences.
- RAG Knowledge Agent - Retrieves real-time product, review, and market information from trusted sources before generating responses.

The solution also includes a **user-friendly dashboard** (built with Streamlit or React) that visualizes comparisons, trends, price movements, and shopping summaries.

LangFlow is used to orchestrate agent workflows, while watsonx.ai ensures scalable model deployment, embeddings, and AI governance. Granite models power reasoning, summarization, and personalization.

Overall, this solution delivers an **end-to-end intelligent Intelligent Shopping Agent** that act as a smart shopping companion that helps buyers make informed and confident decisions besides recommending products.

Technology Used -

- **Langflow platform** - for designing and orchestrating multi-agent workflows visually

- **IBM Grainte model** - for eg - ibm-Granite-4-h-small - for natural language understanding, reasoning, and personalization

- **IBM Cloud** - for providing the infrastructure to build, deploy, and scale the Intelligent Shopping Agent efficiently and enabling a **reliable, secure, and scalable environment** for deploying this Agent

- **RAG -** for fetching **real data first, then generate intelligent answers**, making the Intelligent Shopping Agent more trustworthy and effective.

- **IBM Watsonx.ai -** for model deployment, embeddings, and AI governance

Langflow component Used -

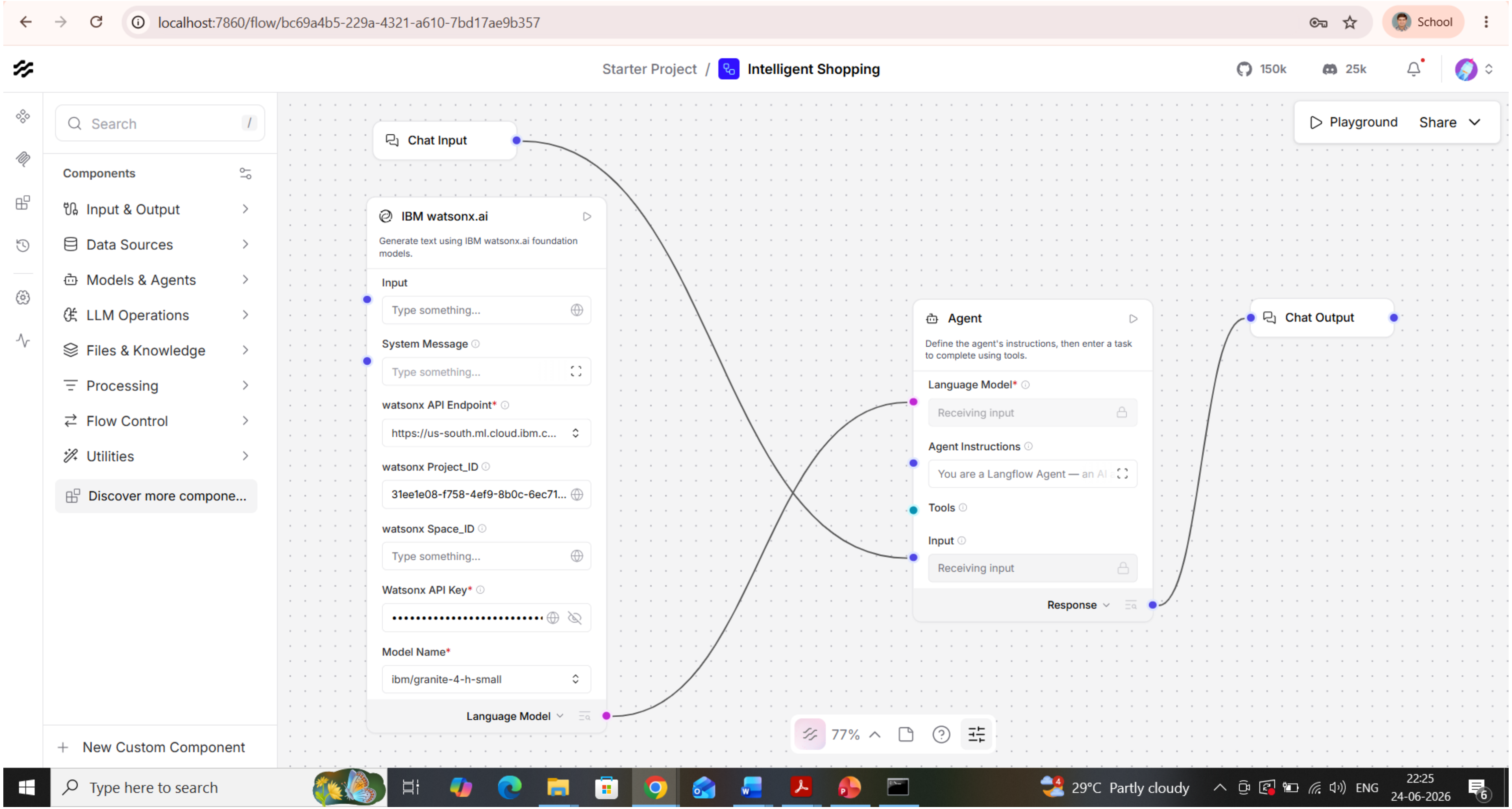
1. **Chat Input** - Interface where users enter their text queries, voice queries, images (e.g., a photo of a product), or text descriptions.

2. **IBM watsonx AI Agent** - Processes user input using granite models to recommend products and helps buyers make informed and confident decisions.

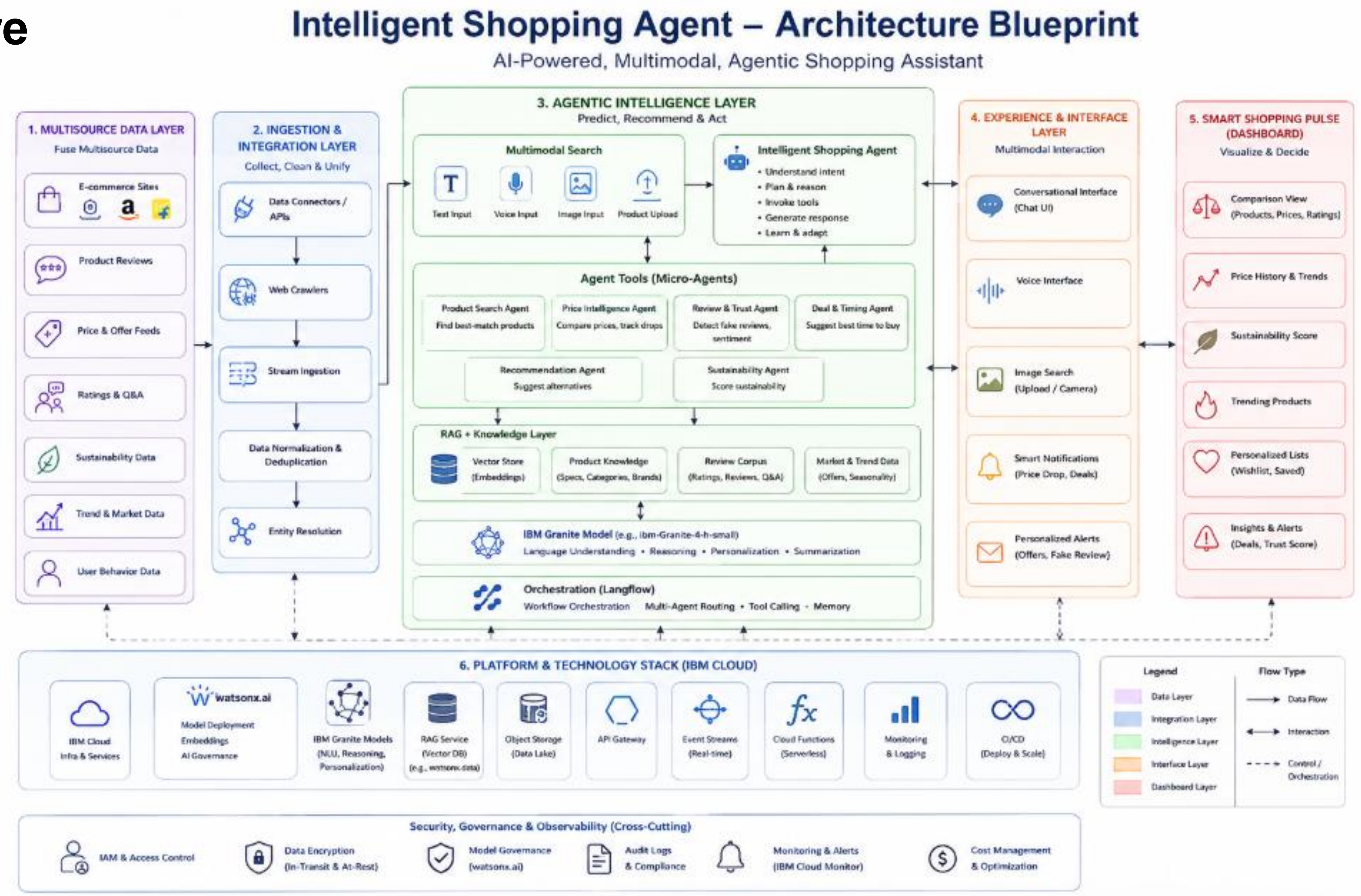
3. **Name of IBM Granite model -** Granite-4-h-small → Fast, efficient for real-time recommendations.

4. **Chat output -** Displays AI-generated responses based on real-time trends, personalized needs, or contextual factors like urgency, budget, or sustainability.

Langflow workflow -



Architecture Blueprint

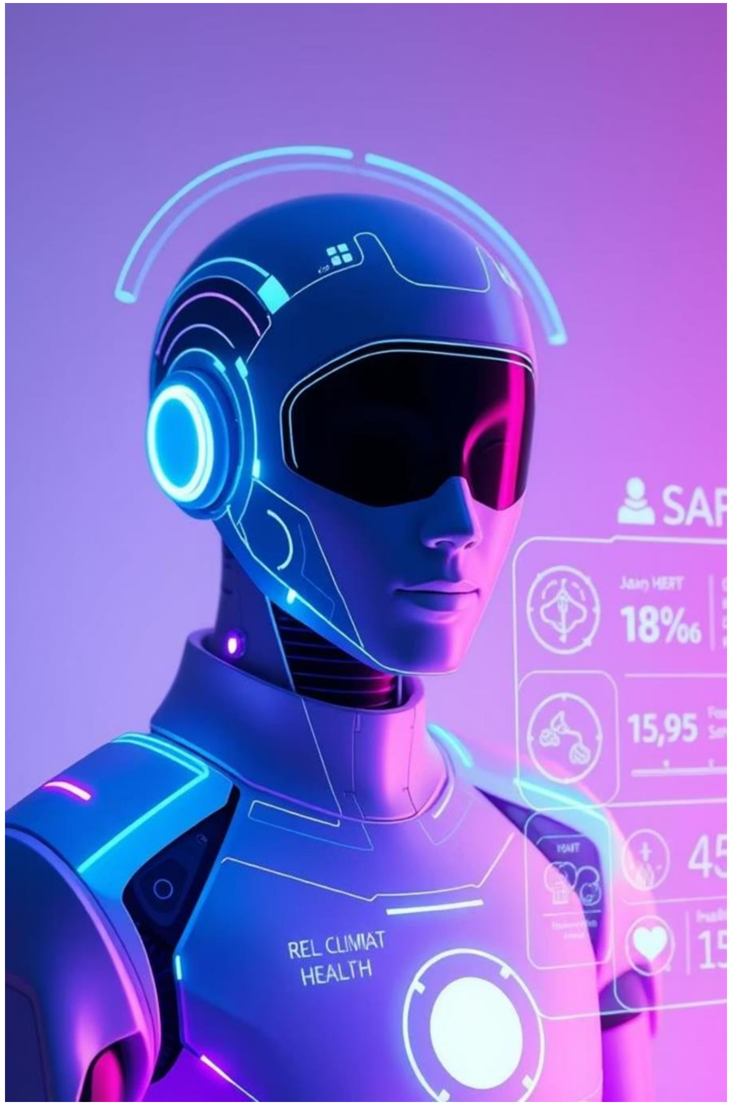


Role of Agentic AI in the solution

Role of Agentic AI in Intelligent Shopping Agent

- **Agentic AI** enables the Agent to function as an intelligent, autonomous system rather than a simple chatbot.
- Using **IBM watsonx.ai** and **IBM Granite models**, it coordinates multiple specialized agents to deliver personalized shopping guidance.
- Each agent performs a specific roles - such as retrieving required data, generating, etc.
- Agentic AI also brings **context awareness**, understanding user details like age, health conditions, and fitness goals to generate tailored recommendations.
- It supports **real-time adaptation**, continuously updating suggestions based on user inputs.
- Additionally, agents communicate with each other, enabling better decision-making and improved outcomes.

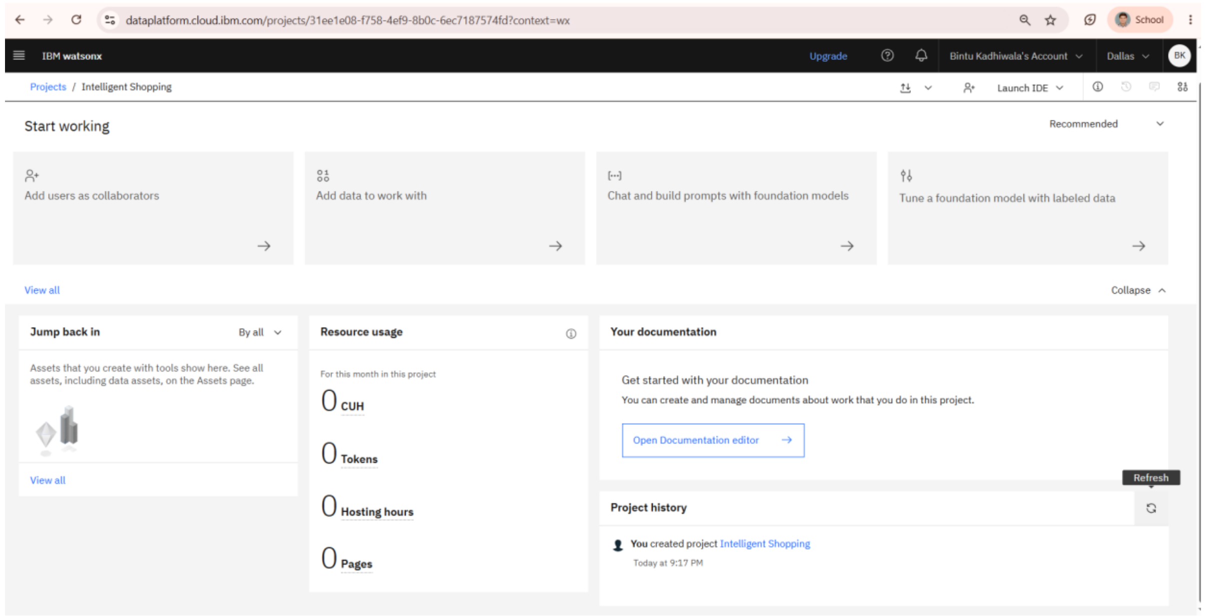
In gist, Agentic AI makes the system proactive, personalized, and goal-driven, acting like a **Intelligent Shopping Agent**.



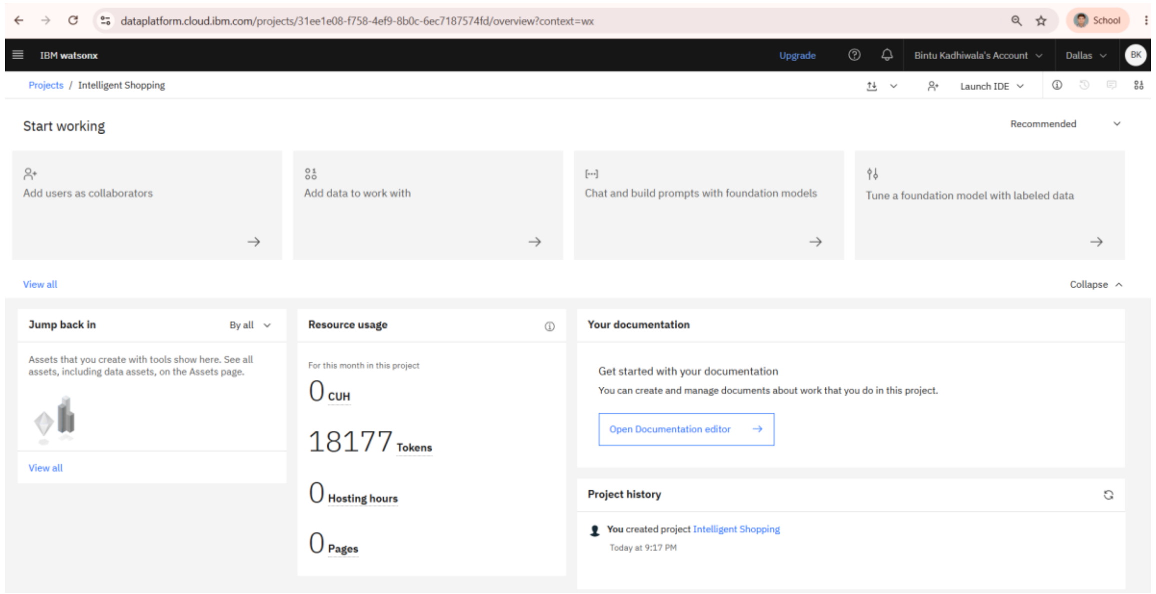
Token Usage

18,177 tokens are used

Before Prompt



After the Prompt



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Novelty and Uniqueness :

The proposed **Intelligent Shopping Agent** stands out by combining **LangFlow, IBM watsonx.ai,** and **Granite Models** into a single intelligent system.

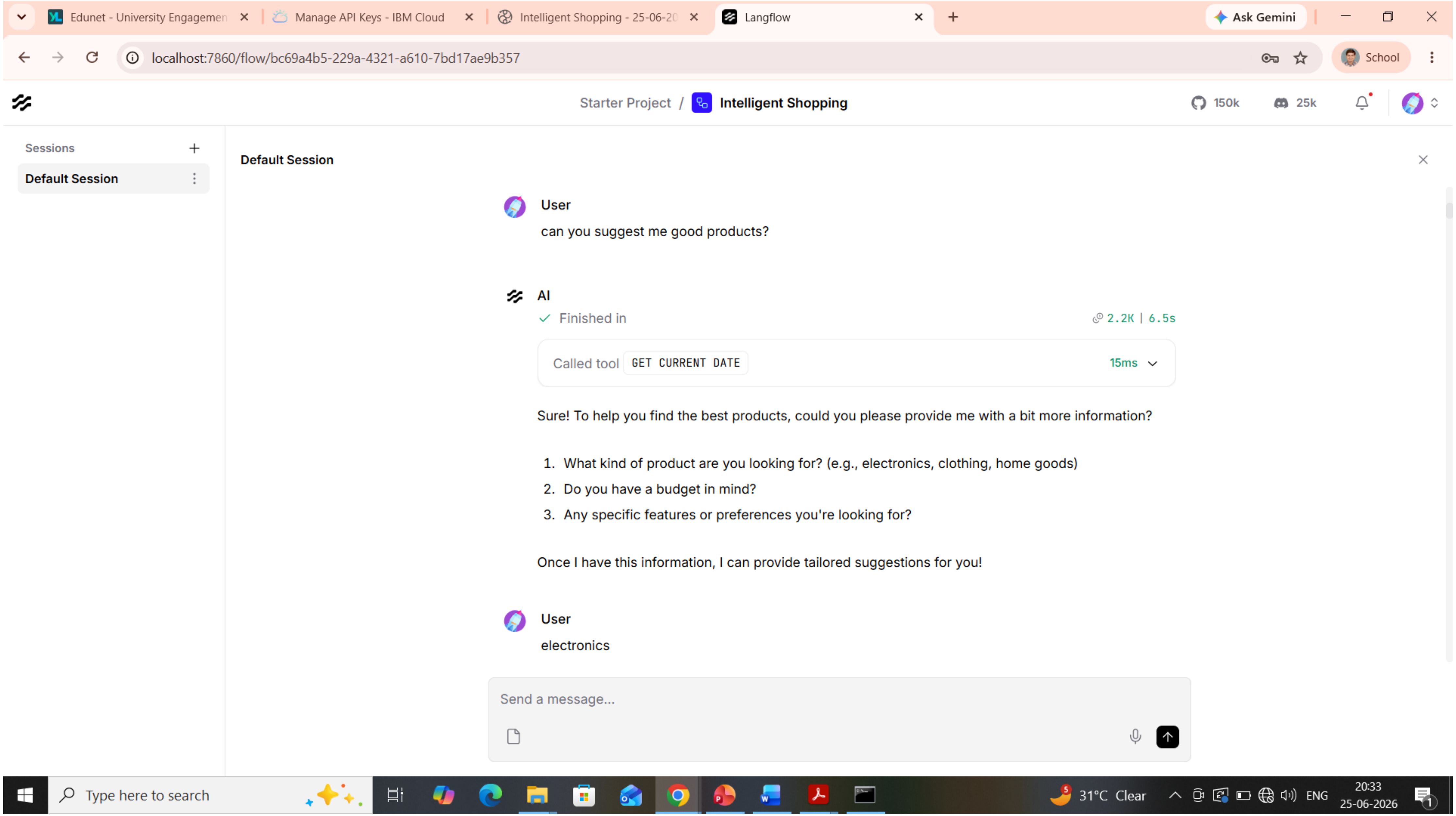
From Search Engine to Autonomous Shopping Agent - It actively assists decision-making throughout the purchasing journey besides responding to users' queries.
For example, "I am working as a professor at computer engineering department. can you suggest me which company laptop should i purchase dell laptop or hp laptop?"

Voice-based/Textual description-based discovery with Agentic Reasoning - It accepts voice and natural language descriptions and combines them with LLM reasoning to identify user intent.
For example, "I am working as a professor at computer engineering department. can you suggest me which company laptop should i purchase dell laptop or hp laptop?"

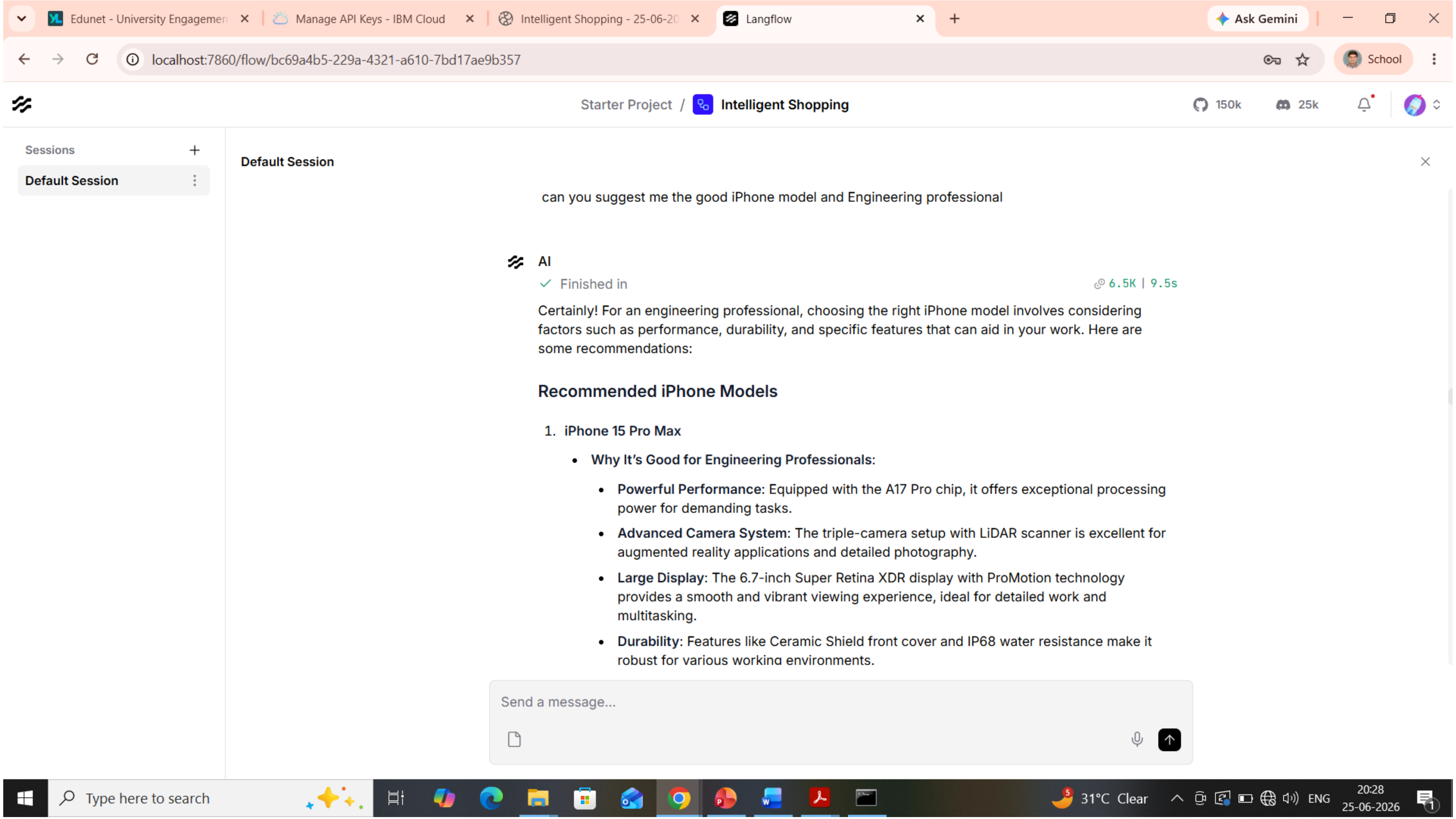
Create a Predictive and Agentic Layer - Suggest purchase timing (e.g., discounts, seasonal offers), recommend alternatives, and alert users about price drops or fake reviews.
For example, "suggest me the proper timings for purchase in the laptops of Dell company"

This makes the solution a **smart, adaptive, and proactive** **Intelligent Shopping Agent**.

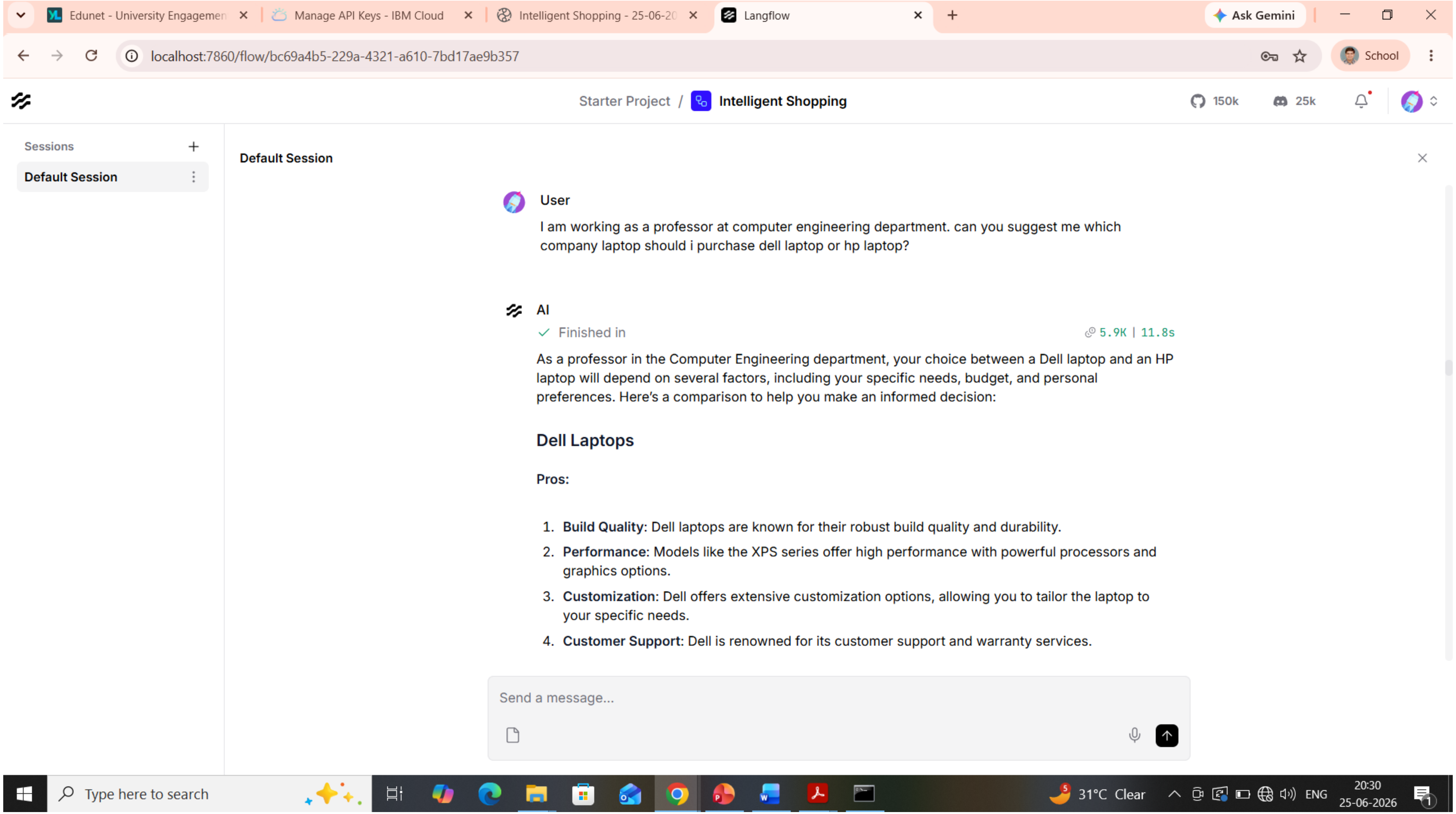
Output Screenshots :



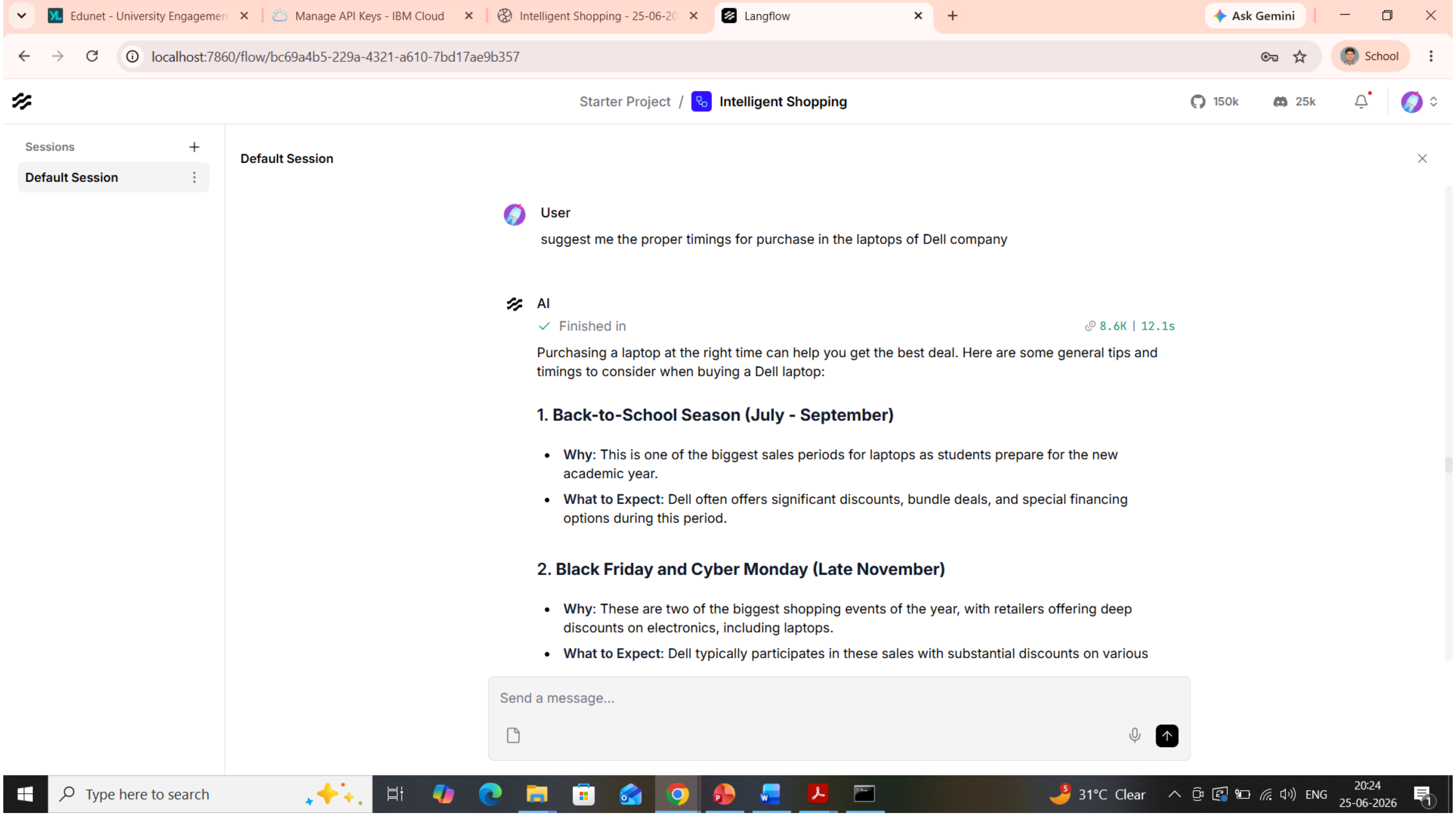
Output Screenshots :



Output Screenshots :



Output Screenshots :



Git Hub Link :

Working Github URL -

https://github.com/bintukadhiwala/Intelligent-Shopping-Agent

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Future Scope :

1. **Integration of Trust and Authenticity Analyzer :** Help users identify reliable products and sellers by generating a "Trust Score" based on reviews, seller reputation, return rates, and warranty coverage, highlighting verified customer experiences, analyzing seller credibility and fulfillment history, generating flag potentially counterfeit products, etc. This feature reduces purchase risk and increases buyer confidence.
2. **Budget Optimization and Cart Intelligence :** Help users maximize value within a budget by building optimal product bundles, suggesting cheaper alternatives with similar quality, etc. For example, a user with a ₹20,000 budget for a gaming setup receives an optimized combination of monitor, keyboard, mouse, and headset. This feature increases customer satisfaction and purchasing efficiency.
3. **Post-Purchase Assistant** : Continue assisting users after the purchase by tracking delivery status, monitoring return and warranty deadlines, recommending compatible accessories after purchase, suggesting setup guides and tutorials (if any), notifying users about recalls or product issues, etc. This feature extends engagement beyond checkout and creates a complete shopping lifecycle assistant.

Certificates Earned :

credly certificate :

In recognition of the commitment to achieve professional excellence

Generative AI in Action
IBM SkillsBuild

Bintu Kadhiwala

Has successfully satisfied the requirements for:

Generative AI in Action

Issued on: Jun 22, 2026
Issued by: IBM SkillsBuild

Verify: https://www.credly.com/badges/92534d40-a367-4dd9-96ab-558d13dbbb43

IBM

Certificates Earned :

IBM BOB Certificate

IBM **SkillsBuild** Completion Certificate

This certificate is presented to

Bintu Kadhiwala

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 23 Jun 2026 (GMT)

Learning hours: 30 mins

Certificates Earned :

Getting Started with Cybersecurity Certificate

In recognition of the commitment to achieve professional excellence

Getting Started with Cybersecurity

IBM SkillsBuild

IBM

Bintu Kadhiwala

Has successfully satisfied the requirements for:

Getting Started with Cybersecurity

Issued on: Jun 24, 2026

Issued by: IBM SkillsBuild

Verify: https://www.credly.com/badges/f832369d-b382-4d18-907a-ec4f1a6be96a

IBM

Certificates Earned :

Exploring Quantum Computing Certificate

IBM **SkillsBuild** Completion Certificate

This certificate is presented to

Bintu Kadhiwala

for the completion of

Exploring Quantum Computing

(ALM-COURSE_3825253)

According to the Adobe Learning Manager system of record

Completion date: 25 Jun 2026 (GMT)

Learning hours: 50 mins

Thank You!

Thank you for your time and interest.